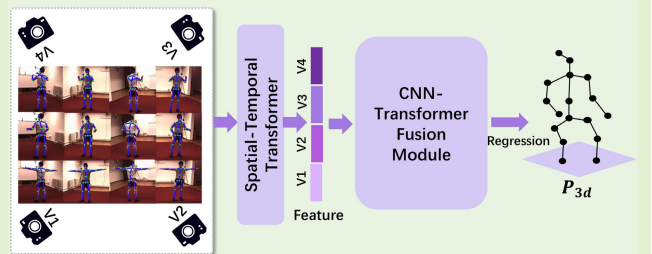


ECTFormer: Efficient CNN-Transformer Network for Uncalibrated Multiview 3-D Human Pose Estimation

Jucheng Song^{1b}, Xu Yang^{1b}, *Member, IEEE*, Yapeng Wang^{1b}, *Member, IEEE*,
Jie Zhang^{1b}, *Member, IEEE*, and Sio-Kei Im^{1b}, *Senior Member, IEEE*

Abstract—Single-view camera sensors suffer from inherent depth blurring issues that hinder progress in 3-D human pose estimation (HPE), sparking widespread research interest in multiview camera sensor systems. However, existing methods typically rely on complex camera calibration processes and are sensitive to dynamic environments. To address these limitations, we propose ECTFormer, an innovative calibration-free multiview 3-D HPE framework that seamlessly combines Transformer-based spatiotemporal modeling with convolutional neural network (CNN)-based local feature extraction. The primary contents of this article are: Firstly, we introduce a hierarchical multiview spatiotemporal feature extraction network. This network avoids interference between noise from different viewpoints through hierarchical learning and employs a transformer to capture spatio-temporal features within views for subsequent fusion. And then, we design a CNN-transformer fusion module (CTFM) that efficiently aggregates multiview features, enabling accurate 3-D pose regression. Extensive experiments on a public 3-D human pose benchmark demonstrate that our approach attains superior performance without relying on calibration. For real-world environments equipped with dynamic camera sensors, ECTFormer can efficiently perform HPE.

Index Terms—Convolutional neural network (CNN), multiview 3-D human pose estimation (HPE), transformer, uncalibrated camera.



I. INTRODUCTION

IN COMPUTER vision research, 3-D human pose estimation (HPE) is a foundational task dedicated to inferring the positions of anatomical joints to model the human body. It serves a wide range of purposes, spanning from motion recognition and immersive virtual reality experiences to medical diagnostics and digital content generation for entertainment media.

Research in 3-D HPE is conventionally split into single-view and multiview approaches. Techniques falling under the former category endeavor to reconstruct 3-D poses from a single visual input [1], [2], [3], [4], [5], [6], [7], [8], [9], be it

an image or a video sequence. However, due to the fixation defect of depth ambiguity in a single-view angle, most of the methods try to use attention to extract global pose features as well as a graph convolutional network (GCN) network to extract local joint features for modeling, which leads to the parameters of the model becoming bigger by constantly updating the model structure, but the improvement of the accuracy is negligible. The accuracy again decreases drastically in the face of mostly occluded environments.

Compared to single-view methods, multiview 3-D HPE methods [10], [11], [12], [13], [14] incorporate observations from additional viewpoints. These methods significantly reduce depth ambiguity and enhance 3-D HPE robustness, particularly in challenging scenarios involving severe occlusions or complex body configurations. As shown in Fig. 1, current state-of-the-art multiview techniques primarily rely on two key strategies: geometric triangulation and cross-view feature aggregation, both of which operate under the assumption of known camera parameters. The requirement for precise camera synchronization and meticulous calibration procedures introduces significant implementation complexity and cost barriers. Furthermore, the performance of such systems tends to degrade in dynamic settings where camera positions may vary or calibration parameters drift over time. These limitations have prompted growing interest in developing more flexible

Received 18 November 2025; revised 6 January 2026; accepted 20 January 2026. Date of publication 3 February 2026; date of current version 16 March 2026. This work was supported in part by the Macao Science and Technology Development Fund (FDCT-MOST) Joint under Project 0018/2025/AMJ and in part by Macao Polytechnic University under Grant RP/FCA-01/2025, and under submission control (code: fca.a4da.bbd3.e). The associate editor coordinating the review of this article and approving it for publication was Dr. Guofa Li. (*Corresponding author: Xu Yang.*)

The authors are with the Faculty of Applied Sciences, Macao Polytechnic University, Macau, China (e-mail: P2424452@mpu.edu.mo; xuyang@mpu.edu.mo; yapengwang@mpu.edu.mo; jpeter.zhang@mpu.edu.mo; marcusim@mpu.edu.mo).

Data is available on-line at <https://github.com/songxiaoguo/ECTFormer>

Digital Object Identifier 10.1109/JSEN.2026.3658078

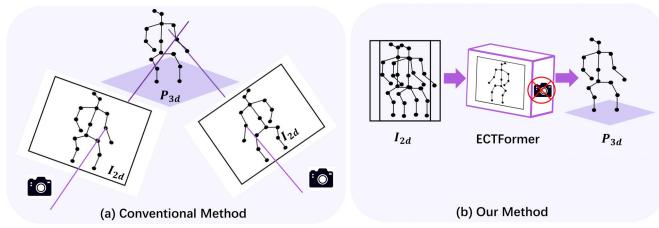


Fig. 1. Comparison between ECTFormer and traditional multiview 3-D HPE methods [11], [12]. (a) Traditional multiview 3-D HPE methods compute 3-D poses P_{3d} through 2-D pose I_{2d} geometry based on camera parameters. (b) Fusion module for 3-D pose estimation eliminates the need for camera parameters, successfully omitting camera calibration operations and enhancing the model's adaptability to a wider range of scenarios.

multiview approaches that can maintain high estimation accuracy while reducing dependency on perfect calibration and synchronization. Recent works have begun exploring deep learning-based solutions to mitigate these constraints.

Therefore, parameter-free multiview methods [15], [16], [17], [18] have emerged as a more practical alternative, eliminating the need for precise camera calibration and synchronization. These approaches are inherently more adaptable to real-world environments, where camera configurations may be dynamic or imperfect. However, achieving such flexibility demands substantial computational resources, requiring top-performing methods to employ intricate network designs for extracting and fusing features across different views. The intricate design of these models, often incorporating attention mechanisms, graph neural networks, or transformer-based modules, results in a significant increase in both parameter size and computational overhead.

To solve these issues, we design a new camera-parameter-free multiview 3-D HPE method. Unlike traditional methods, the ECTFormer does not need relabeling or retraining when the camera configuration or environment changes, thus greatly simplifying the entire process. This inherent adaptability makes our solution particularly practical and efficient in real-world applications. Specifically, firstly, we introduced a transformer architecture to study the spatiotemporal characteristics of human joints from multiple perspectives. Then, a fusion module combining convolutional neural network (CNN) and Transformer is designed to integrate these features. Unlike conventional fusion methods, we utilize a CNN network to extract local features and a transformer to extract fused global features, in order to generate accurate 3-D pose estimates in all views. This greatly simplifies the overall network structure while reducing model parameters. Notably, our approach achieves optimal accuracy without relying on complex model architectures or complicated loss functions. The system effectively addresses the depth ambiguity inherent in single-view 3-D attitude estimation while completely eliminating the recalibration requirements typically found in dynamic multiview environments.

This work introduces the following novel contributions.

- 1) We propose a novel camera-free parametric approach for multiview 3-D HPE, thereby circumventing complex camera calibration procedures. This method efficiently

captures pose-related information across views through the synergistic interaction of spatial transformers, temporal transformers, and convolutional neural networks, comprehensively integrating these features to accurately estimate 3-D human poses.

- 2) We employ a spatial-temporal transformer for coarse feature extraction and utilize an additional spatial-temporal transformer for fused feature extraction to capture long-range cross-view correlations and parallax cues, enabling robust pose estimation.
- 3) To account for pose variations in different view-points, we developed a CNN-Transformer fusion module (CTFM) to efficiently aggregate multiview features to assist in 3-D pose regression.
- 4) Rigorous cross-dataset validation on established 3-D HPE benchmarks substantiates the efficacy of our framework. The proposed method achieves competitive performance accuracy while eliminating dependency on extrinsic camera parameters or intricate loss function engineering.

II. RELATED WORK

A. Single-View Single-Frame Methods

Single-view HPE suffers from the problem of maladaptive depth ambiguity. To address this, the spatial dependencies between human joints, both on a global scale and within local contexts, are frequently utilized to impose constraints on their predicted positions and mitigate depth-related uncertainties. Firstly, Tsai and Luo [1] designed an end-to-end method that can predict a 3-D spatial skeleton from a figure by utilizing the context information in RGB images. And Zhao et al. [19] first used a GCN for capturing global-local feature relationships. However, traditional joint modeling approaches rely on handcrafting, which often results in limited representation and generalization. To overcome this, Yan et al. [20] designed a dynamic skeleton network that automatically learns global-local feature relationships of joints, enhancing feature learning and generalization capabilities. Additionally, Yu et al. [21] introduced the adaptive graph convolutional network that uses graphs to model spatiotemporal structures for 3-D HPE through individually connected backward local joint features. And Li et al. [22] designed a graph convolution network based on an MLP network for 3-D HPE with a joint global and local architecture. To enhance the generalizability of the network, Gong et al. [23] designed a method to generate synthetic data, but this method did not fully utilize temporal features. However, in the single-view domain, the single frame cannot solve the problems caused by depth blur and occlusion environment, while temporal feature information can alleviate the problem of inaccurate pose estimation during occlusion, so there are many studies centered around the multiframe environment.

B. Single-View Multiframe Methods

As transformers have shown great potential in computer vision, most research has focused on using transformers for spatiotemporal features in 3-D HPE. Zheng et al. [4] firstly designed the attention-based approach, demonstrating the strong capabilities of the transformer in 3-D HPE.

Subsequently, Shan et al. [24] proposed P-STMO, which uses masks to randomly obscure human joints, thereby improving performance. Tang et al. [25] disentangled the conventional Transformer architecture into isolated spatial and temporal modules to enhance modeling efficacy. Li et al. [26] designed a multioutput attention network that yields several feasible 3-D pose hypotheses, resulting in improved accuracy. Zhu et al. [27] proposed for the first time the use of a parallel-structured transformer to engage in 3-D HPE owing to its capacity to represent the spatial-temporal capabilities between skeletal joints comprehensively. Sun et al. [28] similarly used a two-stream architecture for feature extraction, with the introduction of the two depth-consistency losses, which establish a more robust and reliable cue to accurately estimate the pose by transferring the relative depth-consistency to the relative localization through the precise ordering of the joints.

Compared with the transformer architecture, current research based on Mamba is also emerging in an endless stream. Firstly, Huang et al. [29] first proposed the task of 3-D HPE based on the mamba network, in which a bidirectional global-local spatio-temporal SSM module was designed, which holistically captures human joint interdependencies within individual frames and temporal dependencies between frames. Combining the efficient performance of Mamba and the dual-stream architecture, Zhang et al. [30] proposed a novel inattention-free hybrid spatiotemporal structure. Through the dual-stream structure, GCN models local joint features, while Mamba captures global joint context. Features from Mamba and GCN are adaptively fused to dynamically integrate to enhance pose estimation accuracy.

While existing methods primarily focus on analyzing spatiotemporal features and joint-level local/global information, they remain fundamentally limited by the inherent depth ambiguity problem in single-view scenarios. These limitations persist because monocular approaches cannot fully resolve the perspective-dependent nature of 3-D pose estimation. Multiview methods address this critical challenge by leveraging complementary information from multiple perspectives, where the geometric constraints provided by additional viewpoints effectively mitigate depth ambiguity through cross-view consistency.

C. Multiview 3-D HPE Conventional Methods

Multiview single people 3-D HPE generally utilizes camera position relationships for modeling, constructs polar geometric constraints, and utilizes feature complements from other viewpoints to reduce the effect of occlusion. Gao and Wang [10] proposed a multiframe temporal fusion module that utilizes the motion context of consecutive frames and refinement of the heatmap, and dynamically adjusts view weights based on the joint confidence scores to achieve a more robust key-point detection. Isakov et al. [11] employed triangulation for 3-D HPE. Remelli et al. [31] mapped multiview features into a volumetric representation for 3-D coordinate regression. Additionally, Liao et al. [32] incorporated camera parameters into positional codes to guide the model in capturing multiview feature relationships.

Compared to the conventional triangulation methods in the multiview single-person domain, the multiview multi-person domain has more methods. First, Lin and Lee [12] and Song et al. [14] introduced a geometry-aware stereo scanning framework that codesigns inter-view feature integration with monocular 3-D pose recovery. Wang et al. [33] advocated a new transformer network for direct regression of multiperson 3-D poses, obviating intermediate representations. Tu et al. [34] and Ye et al. [35] conducted processing in volumetric space, consolidating multiview features within a 3D voxel grid for direct skeletal coordinate regression.

Although these methods have achieved extremely high accuracy in attitude estimation, the complex camera calibration impedes real-world deployment and exhibits limited adaptability to camera configurations with dynamic variations. Therefore, it becomes more meaningful to obtain a method that does not require camera calibration.

D. Multiview 3-D HPE Camera-Parameter-Free Methods

Compared to conventional methods that require camera parameters, parameter-free methods are more suitable for a wider range of scenarios. Therefore, extensive research is currently being conducted on multiperspective parameter-free methods.

Jiang et al. [36] introduced a probabilistic triangulation module that models camera poses using probability distributions. Gordon et al. [15] developed a viewpoint-independent representation of the skeleton, modeling the human body based on skeleton lengths and angles. Zhang et al. [18] introduced a cross-view reasoning framework leveraging adaptive graph transformers to dynamically aggregate discriminative semantic attributes of human joints across perspectives, enhancing 3-D localization fidelity. Zhang et al. [37] also proposed a new attention mechanism to extract different types of features, and combined it with a cross-level fusion module to fuse the poses of all views, to obtain spatio-temporal dependent features between human joints.

However, we found that the network structure of the above method becomes increasingly complex when integrating multiview models, leading to excessive use of computing resources and not proportional to the actual prediction accuracy. Therefore, based on real-life scenarios, we propose a simpler network structure to achieve better performance, which becomes more practical.

III. METHOD

As illustrated in the pipeline of the introduced approach depicted in Fig. 2, the overall procedure can be summarized as follows: an image list $I = (I_i)_{i=1}^{W \times H \times 3}$, where W and H represent the image height and width. The 2-D pose $P_{2-D} \in \mathbb{R}^{T \times V \times J \times 2}$ in every frame is then extracted using CPN [38], where T represents the frames and V represents the views. Following this, the multiview P_{2-D} data is passed into the ECTFormer to obtain the canonical 3-D pose $P_{3-D} \in \mathbb{R}^{T \times J \times 3}$.

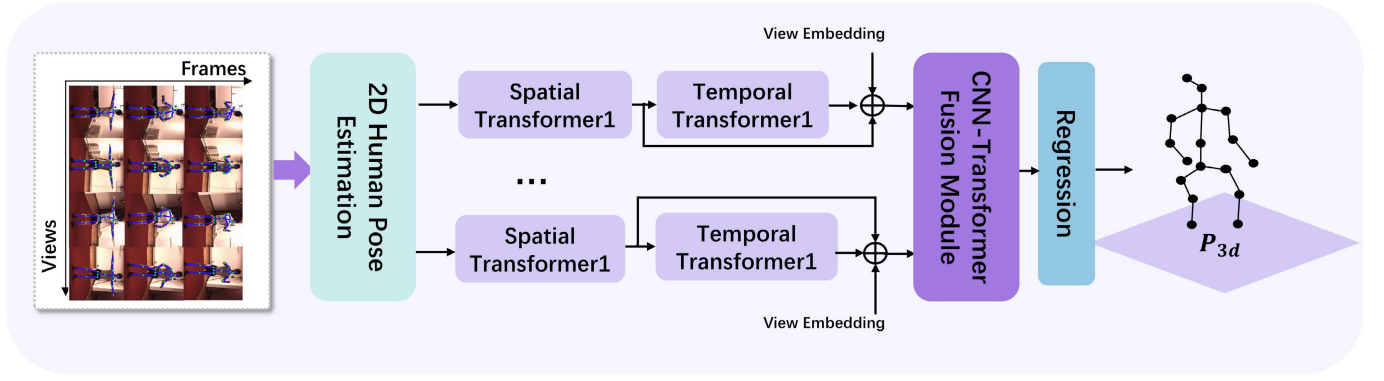


Fig. 2. Overview of ECTFormer. ECTFormer adopts a two-stage network architecture. The first stage employs a transformer-based architecture to independently extract spatio-temporal features from each view. The second stage integrates multiview features through a fusion module, capturing coherent and unique characteristics across views to achieve direct regression of 3-D pose. The overall framework implicitly learns multiview geometric relationships by combining Transformers with CNNs, eliminating the need for camera parameters during pose estimation.

A. 2-D Human Pose Extractor

In established 3-D pose estimation techniques, the approaches are generally classified into two main types. The first category encompasses the methods that regress 3-D poses directly from the image. In comparison, the second approach employs a sequential framework: initially estimating 2-D joint positions before estimating 3-D joint positions. ECTFormer adopts the second type of method, using the CPN [38] conventional pose estimator to extract 2-D poses from multiple perspectives. The specific operation is as follows:

$$F_{2-D} : I^{W \times H \times 3} \longrightarrow \mathbb{R}^{T \times V \times J \times 2} \quad (1)$$

where J denotes the quantity of joints, H and W indicate the vertical and horizontal dimensions of each feature map, and CPN_{2-D} refers to a 2-D HPE network

$$P_{2-D} = CPN_{2-D}(I) \in \mathbb{R}^{T \times V \times J \times 2}. \quad (2)$$

For the multiview approach, we perform 2-D pose estimation separately for each view to obtain 2-D poses, which are used in the 2D-to-3D stage.

B. Views Relationship Extractor

After obtaining the 2-D pose $P_{2-D} \in \mathbb{R}^{T \times V \times J \times 2}$, ECTFormer utilizes the spatio-temporal features to regress the 3-D pose. Next, the designed Transformer architecture is used to extract spatio-temporal features, and linear projection is initially employed to transform the 2-D pose into a high-dimensional feature space

$$P = \text{Linear}(P_{2-D}). \quad (3)$$

Next, we will extract spatio-temporal features from high-dimensional features. The spatio-temporal module of the Transformer architecture used in this article refers to the method [4]. Initially, the spatial module is employed to capture features $S \in \mathbb{R}^{V \times T}$ from 2-D poses, obtaining feature relationships between joints. Then, features with spatial relationships are input into the temporal module for learning, obtaining temporal features $T \in \mathbb{R}^{V \times T}$, and combining spatial features with temporal features. The specific operation is as follows:

$$T = \text{TE}(S), \quad S = \text{SE}(P) \quad (4)$$

where SE corresponds to the spatial feature extraction network, while the TE indicates the temporal feature extraction network. S stands for the resulting spatial representation, and T signifies the resulting temporal representation.

C. CNN-Transformer Fusion Module

In CTFM, the spatial features S and temporal features T of multiple views are efficiently fused to alleviate the problem of depth blurring. To efficiently fuse features, we adopt the CNN combined with the transformer architecture for learning. By leveraging the powerful local feature extraction capability of CNN and integrating it with the global feature extraction of the transformer, we attain favorable outcomes utilizing a compact architecture with a limited parameter count. And through the network, multiview features are directly integrated, avoiding the use of camera parameters. The detailed process is as follows:

$$F' = \text{Cat}(F) = \text{Linear}(S + T + V_{\text{pos}}) \quad (5)$$

$$F''' = \text{Conv2d}(\text{Drop}(F')) \approx \sum_v^V W_v * X_v \quad (6)$$

where w is the weight corresponding to the convolution on the viewpoint axis, and $*$ represents the local weighting along the time or joint dimension

$$F = \text{LN}(F''' + \text{Drop}(\text{Cat}(F'))) \quad (7)$$

where V_{pos} is the encoding matrix for perspectives, and through the linear layer, it maps out the feature matrix $F' \in \mathbb{R}^{V \times T}$ that contains all perspectives. After the comprehensive fusion of local and global features from multiple perspectives, we refine the extracted features F and perform regression to obtain the final predicted 3-D pose $P_{3-D} \in \mathbb{R}^{T \times J \times 3}$. The overall process is as follows:

$$T = \text{TE}(S), \quad S = \text{SE}(F) \quad (8)$$

$$P_{3-D} = \text{Head}(T) \quad (9)$$

where in the encoder component in this context is maintained identically as previously to avoid augmenting the network's learnable weights, and the Head functions as an estimation procedure.

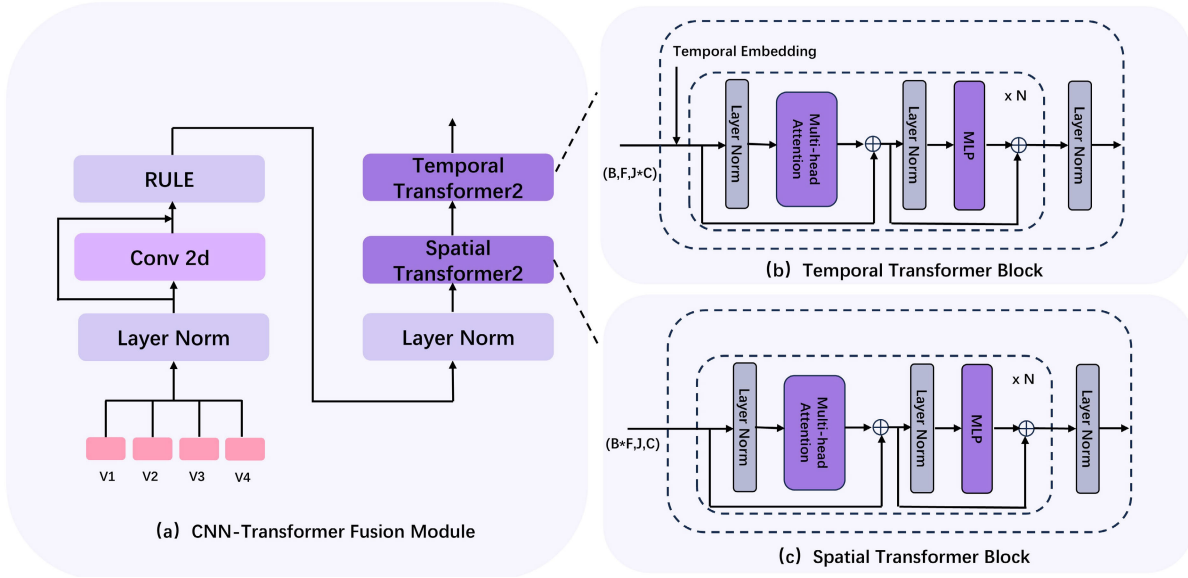


Fig. 3. Overview of CTFM. (a) Architecture of CTFM, which consists of a spatiotemporal module from the Transformer and a CNN. First, we use the CNN for local multiview feature extraction, then fuse the raw features and perform refined global feature extraction through the spatiotemporal module of the Transformer. Throughout the multiview fusion process, we avoid using camera parameters, thereby eliminating the need for camera calibration. (b) Transformer network for processing temporal features, incorporating a temporal encoding matrix. (c) Depicts the transformer network for processing spatial features.

For processes that do not require camera parameters, neural networks are mainly used to learn implicit geometric relationships in multiview features, replacing triangulation that requires camera parameters. For the convenience of understanding, we have introduced triangulation, and the process is as follows:

$$x_{c,j} \sim P_c \cdot X_j \quad (10)$$

where the projection matrix for view c is P_c , where j counts the joints and c the view index, The 2-D position of joint j in view c is given by $x_{c,j}$, while $w_{c,j}$ are parameters learned to modulate the contribution of each view during triangulation, safeguarding against the adverse influence of incorrect 2-D detections. Expanding the above formula into a linear equation can be written as

$$w_j \odot A_j \cdot X_j = 0 \quad (11)$$

$$A_j^* = w_j \odot A_j, \quad A_j^* = U \Sigma V^T \quad (12)$$

where A_j comprises the projection matrix P_c and the learnable parameters $w_{c,j}$, where we annotate the $A_j^* = w_j \odot A_j$ as triangular matrix. For A_j an SVD solution can be performed with the formula $A_j^* = U \Sigma V^T$, where X_j is the last column of the V matrix

$$X_j^* = \arg \min_{X_j} \sum_{c=1}^C w_{c,j} \cdot \|x_{c,j} - P_c \cdot X_j\|^2 \quad (13)$$

where $\|\cdot\|^2$ denotes the L_2 norm. The solution X_j^* as the 3-D pose calculated through triangulation.

D. Spatial-Temporal Transformer Block

In our proposed method, as shown in Fig. 3, we employed two spatiotemporal transformer models. The main difference of the spatiotemporal conversion module is the joint dimension

of the input, where H is the feature input to the spatiotemporal transformer. The entire process is as follows:

$$P = H + T_{\text{pos}} \quad (14)$$

$$Q_p = \text{LN}(P), \quad K_p = \text{LN}(P), \quad V_p = \text{LN}(P) \quad (15)$$

$$X_p = \text{Softmax}\left(Q_p, K_p^T / \sqrt{d}\right) V_p \quad (16)$$

$$S = \text{MLP}(X_p + P) + X_p \quad (17)$$

where T_{pos} is a learnable Temporal location encoding that differentiates Temporal features across viewpoints. It is worth noting that we did not incorporate positional encoding in the spatial transformer module. LN is a linear layer.

E. Loss Function

We first used weighted average per joint position error (WMPJPE), which is a loss function that pays different attention to different joints of the human body when calculating MPJPE. Focus on different weights W

$$L = \frac{1}{W} \sum_{i=0}^w (\theta_i \|J_{3-D} - G_{3-D}\|_2) \quad (18)$$

where J_{3-D} represents the predicted 3-D location of joint i , while G_{3-D} corresponds to its estimated true position, and W indicates the weight coefficient assigned to joint i . And we adopted L_2 loss to reduce the difference between the output and ground truth values, in order to maximize prediction accuracy.

IV. EXPERIMENT

A. Datasets

Human3.6M [44] serves as the benchmark dataset applied for 3-D HPE, comprising 3.6 M frames recorded via 4 time-synchronized cameras. It contains 15 actions executed by 11 subjects. Adopting the protocol of [4], [18], we designate $S1, S5-S8$ as the train set and $S9, S11$ as the test set.

TABLE I

EXPERIMENTAL RESULTS FOR Human3.6M. WE EMPLOY THE CPN DETECTOR AS THE INPUT TO OUR APPROACH, WHERE T INDICATES THE TOTAL FRAME COUNT, AND GT REPRESENTS THE GROUND-TRUTH 2-D INPUT VALUES USED TO INVESTIGATE THEORETICAL PERFORMANCE LIMITS. OWING TO LIMITED SPACE, ONLY A SUBSET OF REPRESENTATIVE STUDIES IS DISPLAYED. THE HIGHEST PERFORMANCE OUTCOMES ARE HIGHLIGHTED IN RED, WHILE THE SECOND-BEST ARE MARKED IN BLUE

Method(MPJPE)	Published	Dir.	Disc.	Eat	Gree	Phone	Photo	Pose	Purch	Sit	SitD.	Smoke	Wait	WalkD.	Walk	WalkT	Avg.
Single-view methods																	
PoseFormer [4](T=243)	ICCV'21	41.5	44.8	39.8	42.5	46.5	51.6	42.1	42.0	53.3	60.7	45.5	43.3	46.1	31.8	32.2	44.3
MixSTE [5](T=243)	CVPR'22	37.6	40.9	37.3	39.7	42.3	49.9	40.1	39.8	51.7	55.0	42.1	39.8	41.0	27.9	27.9	40.9
STCFormer [25](T=243)	CVPR'23	38.4	41.2	36.8	38.0	42.7	50.5	38.7	38.2	52.5	56.8	41.8	38.4	40.2	26.2	27.7	40.5
D3DP [39](T=243)	ICCV'23	37.7	39.9	35.7	38.2	41.9	48.8	39.5	38.3	50.5	53.9	41.6	39.4	39.8	27.4	27.5	40.0
MotionAGFormer [6](T=243)	WACV'24	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	38.4
KTPFormer [40](T=243)	CVPR'24	37.3	39.2	35.9	37.6	42.5	48.2	38.6	39.0	51.4	55.9	41.6	39.0	40.0	27.0	27.4	40.1
PoseMagic [30](T=243)	AAAI'25	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	37.5
PoseMamba [29](T=243)	AAAI'25	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	37.1
Multi-view methods (Camera-parameter-required)																	
Luvizon et al. [41]	IJCV'22	31.0	33.0	41.0	34.0	41.0	37.0	37.0	51.0	56.0	43.0	44.0	37.0	33.0	42.0	32.0	39.0
Bultmann et al. [42]	RSS'21	27.1	29.9	27.0	26.5	31.3	28.9	27.1	29.8	36.5	36.0	30.8	29.3	29.7	27.3	26.3	29.8
Qiu et al. [43]	ICCV'19	24.0	26.7	23.2	24.3	24.8	22.8	24.1	28.6	32.1	26.9	31.0	25.6	25.0	28.1	24.4	26.2
Iskakov et al. [11]	ICCV'19	19.9	20.0	18.9	18.5	20.5	19.4	18.4	22.1	22.5	28.7	21.2	20.8	19.7	22.1	20.2	20.8
Multi-view methods (Camera-parameter-free)																	
FLEX [15](T=27)	ECCV'22	22.0	23.6	24.9	26.7	30.6	35.7	25.1	32.9	29.5	32.5	32.6	26.5	34.7	26.0	27.7	30.2
Zhang et al. [16](T=27)	ICIP'23	26.1	28.2	24.5	26.9	28.7	31.2	26.4	28.2	31.5	33.6	29.4	25.8	28.8	27.4	27.0	28.3
Jiang et al. [36](T=27)	ICCV'23	24.0	25.4	26.6	30.4	32.1	20.1	20.5	36.5	40.1	29.5	27.4	27.6	20.8	24.1	22.0	27.8
SGraFormer [18](T=27)	AAAI'24	26.5	28.3	23.0	25.9	27.2	31.0	25.4	27.2	28.6	33.8	28.6	25.6	30.1	27.1	26.5	27.6
ESMFormer [37](T=27)	Pattern Recognition'25	24.0	27.5	24.8	25.0	28.8	31.3	24.7	27.0	30.6	36.2	28.6	25.3	30.0	24.7	24.0	27.4
Ours(T=27)	/	23.7	27.8	24.6	24.1	27.6	30.0	23.5	26.0	30.0	32.4	27.5	23.9	28.1	23.9	24.1	26.5
FLEX [15](GT,T=27)	ECCV'22	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	22.9
SGraFormer [18](GT,T=27)	AAAI'24	11.7	13.0	10.1	12.1	10.7	13.0	12.1	10.7	10.8	11.9	11.0	11.6	12.8	11.1	12.0	11.7
ESMFormer [37](GT,T=27)	Pattern Recognition'25	16.5	18.3	15.1	17.2	17.0	19.4	16.8	18.7	22.3	17.5	17.3	20.0	15.4	16.8	17.6	17.6
Ours(GT,T=27)	/	11.1	11.8	9.0	11.6	9.3	11.5	11.4	10.7	8.3	9.3	9.8	10.2	11.8	10.0	10.5	10.4
Method(PA-MPJPE)																	
Single-view methods																	
PoseFormer [4](T=243)	ICCV'21	32.5	34.8	32.6	34.6	35.3	39.5	32.1	32.0	42.8	48.5	34.8	32.4	35.3	24.5	26.0	34.6
MixSTE [5](T=243)	CVPR'22	32.0	34.2	31.7	33.7	34.4	39.2	32.0	31.8	42.9	46.9	35.5	32.0	34.4	23.6	25.2	33.9
STCFormer [25](T=243)	CVPR'23	30.4	33.8	31.1	31.7	33.5	39.5	30.8	30.0	41.8	45.8	34.3	30.1	32.8	21.9	23.4	32.7
D3DP [39](T=243)	ICCV'23	27.5	29.4	26.6	27.7	29.2	34.3	27.5	26.2	37.3	39.0	30.3	27.7	28.2	19.6	20.3	28.7
MotionAGFormer [6](T=243)	WACV'24	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	32.5
KTPFormer [40](T=243)	CVPR'24	24.1	26.7	24.2	24.9	27.3	30.6	25.2	23.4	34.1	35.9	28.1	25.3	25.9	17.8	18.8	26.2
PoseMagic [30](T=243)	AAAI'25	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	32.1
PoseMamba [29](T=243)	AAAI'25	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	31.5
Multi-view methods (Camera-parameter-free)																	
SGraFormer [18](T=27)	AAAI'24	20.5	22.6	19.1	20.2	21.0	24.4	20.1	19.7	23.8	27.2	22.3	21.0	22.7	19.4	20.2	21.6
Ours(T=27)	/	18.8	21.6	19.3	18.7	19.9	22.8	18.7	18.3	23.7	25.5	21.7	19.6	21.3	18.0	19.0	20.5
SGraFormer [18](GT,T=27)	AAAI'24	4.0	4.8	3.7	4.0	3.9	5.3	3.9	4.0	5.0	4.9	4.3	4.1	4.8	3.8	3.8	4.3
Ours (GT,T=27)	/	3.1	3.4	3.2	3.2	3.4	3.6	3.1	3.6	3.9	4.1	3.4	3.2	3.6	3.2	3.6	3.4

MPI-INF-3DHP [45] provides a comprehensive dataset for 3-D pose estimation, incorporating indoor/outdoor scenes with 1.3 M+ images from 14 synchronized multiview cameras. It documents 8 activities by 8 subjects, featuring 17-point kinematic annotations per frame. Adhering to the protocol outlined in [18], we utilize thoracic views of S1–S6 as the train set and S7–S8 as the evaluation.

B. Evaluation Indicators

For the H3.6M dataset, the used evaluation measures are MPJPE and P-MPJPE. Mean per joint position error (MPJPE) measures the mean Euclidean distance across all joints from their predicted locations to the corresponding ground-truth positions, without any additional alignment. Procrustes-aligned MPJPE (P-MPJPE) measures the error in joint locations following a Procrustes alignment between the predicted and ground-truth 3-D poses. For the 3DHP dataset, we adhered to the same metric protocol as Human3.6M, while also including other indicators, such as PCK and AUC, referring to method [18].

C. Experiments Details

Our method adopted the PyTorch framework, and then the experiment only used a single RTX A8000 for training for one

day. We configured the optimizer with 0.1 weight decay and initialized the learning rate at $2e^{-4}$. Per-epoch learning rate decay employed a contraction factor ($\alpha = 0.98$). The offline 2-D detector adopted CPN [38]. Both spatial and temporal Transformer encoders contained 4 layers each, optimized via $L2$ loss.

D. Quantitative Results

1) *Results on Human3.6M*: Our approach was benchmarked on Human3.6M, with quantitative outcomes detailed in Table I. This table presents comparative results of two evaluation metrics.

Initially, regarding the MPJPE metric, our approach yielded a decrease of 10.6 mm relative to the top-performing single-view technique, illustrating the efficacy of the multiview approach in alleviating the intrinsic depth ambiguity found in single-view 3-D pose estimation. In addition, when contrasted with various multiview techniques that require camera parameters, our method has poorer accuracy than these methods, but these methods are often limited by their dependence on static camera configuration and lack of adaptability to dynamic environments. In contrast, our method, within a multiview framework without camera parameters, exhibits strong adaptability to different scenes while achieving state-of-the-art accuracy.

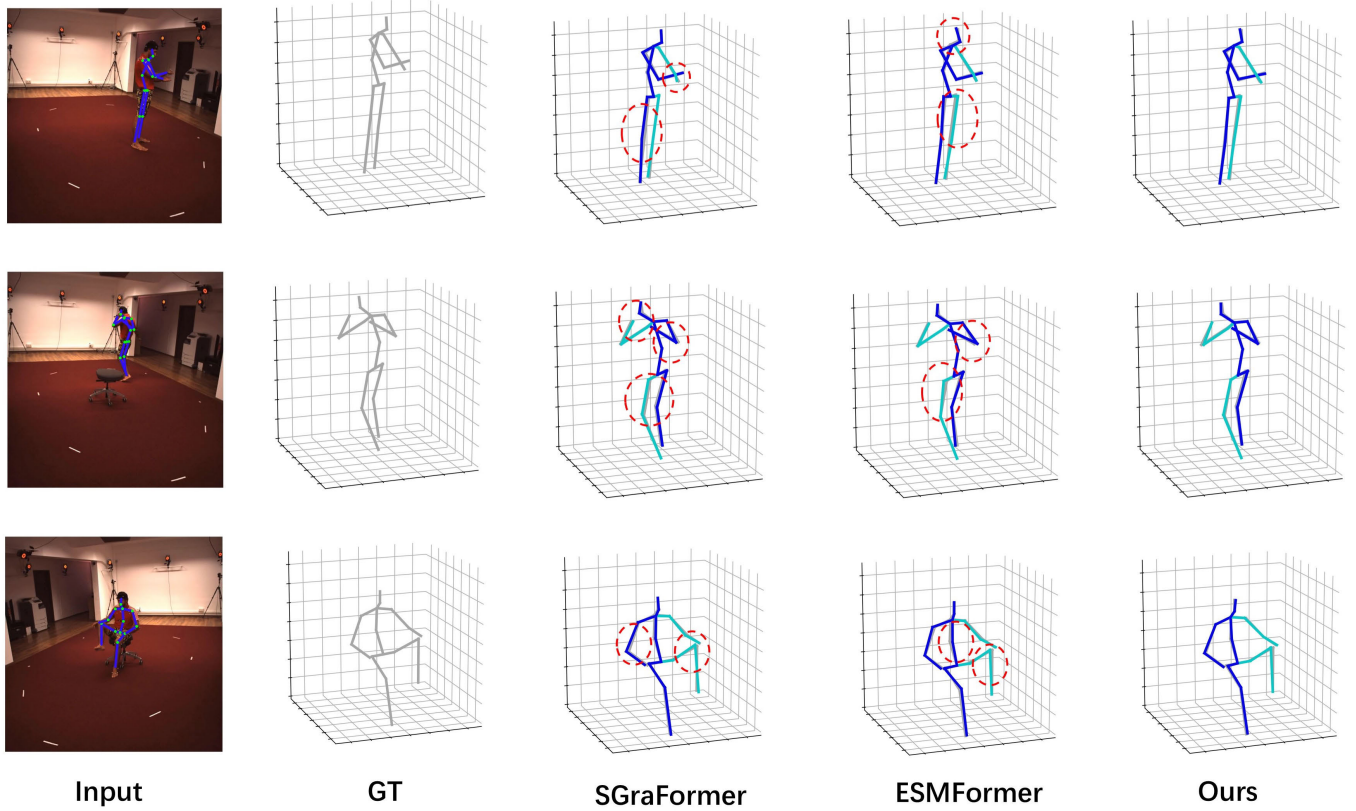


Fig. 4. Qualitative analysis of ECTFormer. On the Human3.6M dataset, we visually compared our method with SGraFormer [18] and ESMFormer [37], where GT denotes ground truth labels. In each experiment, gray indicates the true value of the predicted pose, while red-marked regions indicate the superior performance of ECTFormer compared to others.

Compared to SGraFormer [18], MPJPE has been reduced by 1.1 mm. Compared with ESMFormer [37], MPJPE has been reduced by 0.9 mm. At the same time, when using 2-D ground real attitude input for evaluation, our method achieved an MPJPE of 10.4 mm, a reduction of 1.3 mm compared to SGraFormer [18], and emphasized the key role of accurate 2-D attitude input in improving overall 3-D attitude estimation performance.

Secondly, in the indicator PA-MPJPE, we also compared the methods of single-view and multiview without camera parameters. Our method achieved a PA-MPJPE accuracy of 20.5 mm, which is 11.0 mm less than the single-view method. Compared to the multiview method without camera parameters, it is 1.1 mm less. At the same time, when using true pose input, our method is still 0.9 mm lower than the multiview method. This also verifies the robustness of our method under different indicator requirements and demonstrates its superior performance.

2) Results on 3DHP: This section assessed our method on the 3DHP, and the experimental outcomes are summarized in Table II. Our paper has achieved outstanding performance on multiple metrics, surpassing state-of-the-art levels. Specifically, compared with SGraFormer [18] under the same conditions, our method achieves slightly lower position errors by 0.3 and 0.4 mm, while also improving accuracy and area-under-curve scores by 0.2 and 0.4 percentage points, respectively. When comparing ESMFormer [37], about the

TABLE II
EXPERIMENTAL DATA RESULTS OF 3DHP COMPARED
WITH SINGLE-VIEW AND MULTIVIEW

Method	PCK↑	AUC↑	MPJPE↓	P-MPJPE↓
Single-View Methods				
MixSTE [5]	94.4	66.5	54.9	-
STCFormer [25]	98.4	83.4	24.2	-
D3DP [39]	98.0	79.1	28.1	-
KTPFormer [40]	98.9	84.4	19.2	-
PoseMamba [29]	-	-	17.8	-
Multi-View Methods				
Gholami et al. [46]	-	-	101.5	76.5
Canonpose [47]	77.0	-	104.0	70.3
HMVFormer [48]	98.7	86.8	18.0	13.1
SGraFormer [18]	98.7	90.2	16.9	12.1
ESMFormer [37]	99.5	74.8	21.8	16.7
Ours	98.9	90.6	16.6	11.7

MPJPE and P-MPJPE, our method reduced the metrics by 5.2 and 5.0 mm, respectively, but slightly lower than the ESMFormer [37] method in PCK. We believe this may be due to the specificity of the network structure of this method, but we are much higher than this method in AUC metrics. Taken together, these findings demonstrate that our approach exhibits robust adaptability and delivers outstanding results across diverse datasets.

3) Computational Complexity: We compared the model parameters of our method with those of other approaches, with detailed data from other papers presented in Table III. Compared to ESMFormer [37], our method achieves a 6.0 M

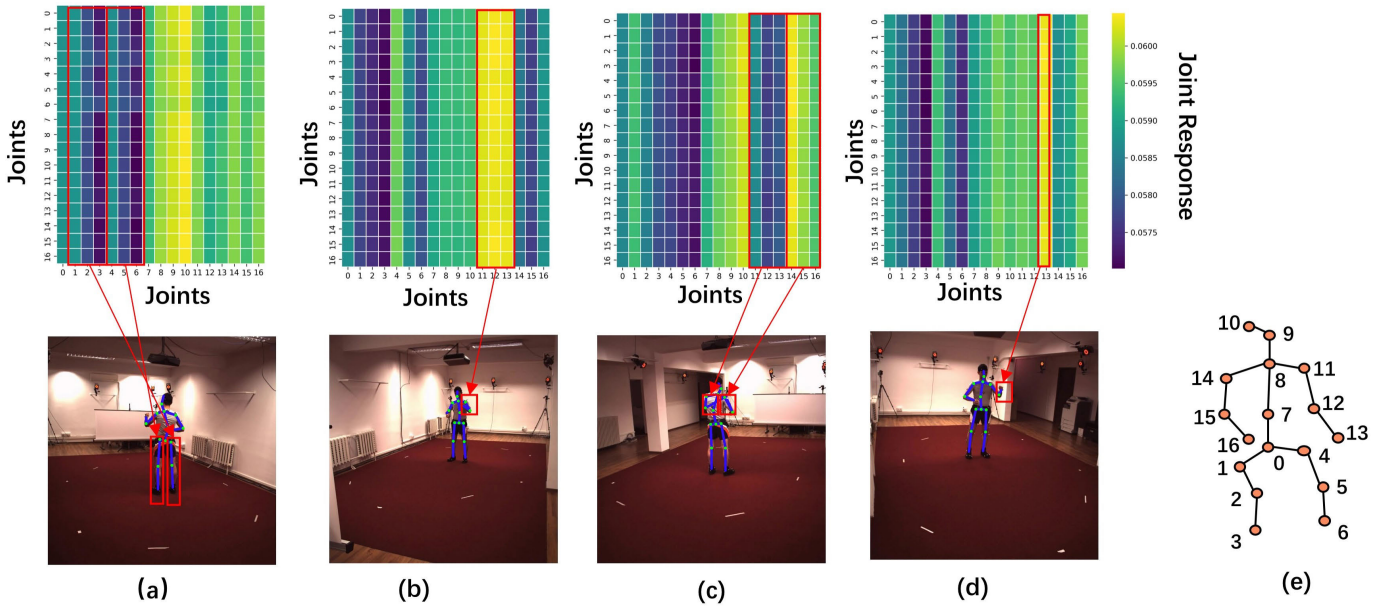


Fig. 5. Visualization of attention relationships among different joints. (a) Joint attention map derived from spatial attention under Viewpoint 1, indicating that joints 1–3 and 4–6 maintain consistent associations with other joints. (b) Joint attention map corresponding to spatial attention under Viewpoint 2, showing that joints 11–13 remain unobstructed, which strengthens their associations with other joints. (c) Joint attention map for spatial attention under viewpoint 3, where the unoccluded joints 14–16 exhibit stronger connectivity with other joints compared to the occluded joints 11–13. (d) Joint attention map for spatial attention under Viewpoint 4, in which the unoccluded joint 13 demonstrates higher connectivity with other joints. (e) Annotation map of the human pose joints.

TABLE III
EXPERIMENTAL RESULTS OF MODEL EFFICIENCY

Methods	F	Params(M)↓	FLOPs(G)↓	MPJPE↓
FLEX [15]	27	70.4	8.5	30.2
Zhang et al. [16]	27	9.7	0.3	28.3
SGraFormer [18]	27	11.4	0.4	27.6
ESMFormer [37]	27	15.6	0.5	27.4
Ours	27	9.6	5.1	26.5

parameter reduction while lowering the MPJPE accuracy by 0.9 mm. Although our algorithm incurs slightly higher FLOPs due to differences in loss function design, we consider this potential drawback in FLOPs justified, given the substantial performance gains achieved.

E. Qualitative Comparisons With SOTA Methods

We display the pose estimation outcomes in Fig. 4 and contrast them with those produced by SGraFormer [18] and ESMFormer [37] on the Human3.6M benchmark. This comparison offers a vivid illustration of the efficacy of our approach. The ECTFormer behaves well when there is severe occlusion or complex postures, highlighting its robustness and accuracy.

F. Ablation Study

1) *Impact of Joint Attention Maps:* Fig. 5 shows attention relationships among different joints. (a) Joint attention map derived from spatial attention under Viewpoint 1, indicating that joints 1–3 and 4–6 maintain consistent associations with other joints. (b) Joint attention map corresponding to spatial attention under Viewpoint 2, showing that joints 11–13 remain

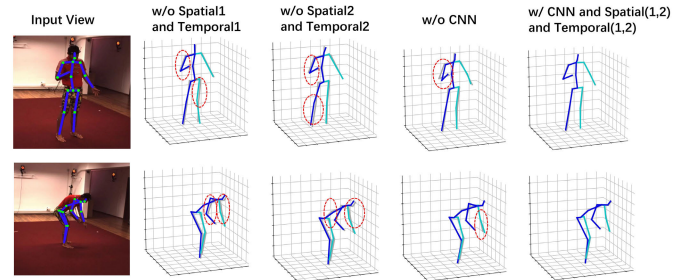


Fig. 6. Quantitative analysis experiment on module ablation. We conducted experiments on the Human3.6M dataset, performing visual comparisons by sequentially removing each module. These comparisons enable an intuitive assessment of the correctness of each module's utilization within the current method.

TABLE IV
EXPERIMENTAL RESULTS OF ABLATION WITH DIFFERENT FRAME RATES. F INDICATES THE TOTAL FRAME

Method	Time(s)	MPJPE↓		P-MPJPE↓	
		2DGT	CPN	2DGT	CPN
f=1	7.4	11.4	28.2	4.0	21.9
f=3	17.1	11.2	27.6	4.1	21.3
f=9	38.6	10.8	27.1	4.0	20.7
f=27	113.5	10.4	26.5	3.4	20.5
f=81	350.9	10.0	26.3	3.0	20.4

unobstructed, which strengthens their associations with other joints. (c) Joint attention map for spatial attention under Viewpoint 3, where the unoccluded joints 14–16 exhibit stronger connectivity with other joints compared to the occluded joints 11–13. (d) Joint attention map for spatial attention under Viewpoint 4, in which the unoccluded joint 13 demonstrates higher connectivity with other joints. (e) Annotation map of the human pose joints. The above experiments validate that

TABLE V
EXPERIMENTAL RESULTS OF ABLATION WITH MULTI VIEWS

Method ($f=27$)	Params(M)	MPJPE↓		P-MPJPE↓	
		2DGT	CPN	2DGT	CPN
View=1	9.5699	34.8	58.0	17.6	46.1
View=2	9.5705	21.4	32.2	10.1	25.5
View=3	9.5711	15.2	29.9	6.4	23.7
View=4	9.5717	10.4	26.5	3.9	20.5

TABLE VI
EXPERIMENTAL RESULTS OF SPATIOTEMPORAL
MODULE ABLATION IN CTFM

Method	MPJPE↓	P-MPJPE↓
w/ Spatial2 and Temporal2	26.5	20.5
w/o Spatial2	28.0	21.3
w/o Temporal2	27.3	21.3
w/o Spatial2 and Temporal2	28.1	21.4

multiview approaches effectively mitigate occlusion effects and achieve more accurate poses by supplementing features from additional perspectives.

2) *Visualization Experiment of Module Ablation*: We visualized ablation experiments for different modules in Fig. 6. The left panel shows the input dataset's 2-D pose visualization. We sequentially removed different modules and visualized the results. For easy comparison, we visualized the ground truth in gray. It is evident that the best performance is achieved when all modules are used together.

3) *Impact of Number of Frames*: Table IV presents the influence of varying frame numbers on model performance. Since frame rate differences do not alter model parameters, we additionally measured the computational time consumption during inference for different frame rates. With a single frame ($f = 1$), the model achieves an accuracy of 28.2 mm. The best accuracy, 26.3 mm, is attained using 81 frames. When using 2-D ground truth input, the accuracy of $f = 27$ is 10.4 mm, and the accuracy of $f = 81$ is 10.0 mm. This observation indicates the importance of the network in the process of temporal feature extraction, and as the frame rate increases, the accuracy tends to improve.

4) *Impact of Number of Views*: Table V presents the effect of different quantities of viewpoints on model performance. As the number of viewpoints increases, our model's performance steadily improves without requiring a significant increase in model parameters. As the quantity of viewpoints augments, our model's performance consistently improves, reaching its peak with 4 viewpoints (the maximum number of viewpoints in the Human3.6M dataset). This indicates that multiview data effectively complements information from other viewpoints and mitigates the depth ambiguity inherent in single-viewpoint scenarios. Additionally, our framework does not require camera parameters and is theoretically capable of accommodating an infinite quantity of viewpoints.

5) *Impact of Number of Modules*: Table VI presents the effect of spatiotemporal modules in CTFM on model effectiveness. Experimental data show that the spatiotemporal module has varying degrees of performance improvement, but when only the spatial module is used, the performance improvement is not significant, while the temporal module has a certain

TABLE VII
ABLATION EXPERIMENT OF POSITION ENCODING. AMONG THEM,
SE IS SPATIAL POSITION ENCODING, TE IS TEMPORAL POSITION
ENCODING, AND VE IS PERSPECTIVE ENCODING

SE	TE	VE	MPJPE↓	P-MPJPE↓
✓	×	×	26.79	20.76
×	✓	×	26.76	20.74
×	×	✓	27.01	20.91
×	×	×	26.99	20.63
✓	✓	✓	26.85	20.54
✓	✓	×	26.53	20.49
✓	×	✓	26.50	20.46
×	✓	✓	26.46	20.46

TABLE VIII
LAYER ABLATION EXPERIMENTS FOR TEMPORAL MODELING
STRATEGIES WERE CONDUCTED TO VALIDATE THE IMPACT
OF LAYER DEPTH ON OVERALL PERFORMANCE. EVALUATION
METRICS INCLUDED CONVENTIONAL MPJPE AND P-MPJPE,
SUPPLEMENTED BY PARAMETER COMPARISONS
FOR EACH LAYER REMOVED

Temporal Layer	Params(M)	MPJPE↓	P-MPJPE↓
Depth=1	3.72	27.91	20.92
Depth=2	6.09	27.09	20.75
Depth=3	8.47	26.90	20.58
Depth=4	9.64	26.50	20.44

TABLE IX
ALL MODULE ABLATION EXPERIMENTS WERE CONDUCTED BASED ON
EACH MODULE'S PARAMETER CONFIGURATION AND ITS IMPACT ON
OVERALL PERFORMANCE. EVALUATION METRICS INCLUDED
TRADITIONAL MPJPE AND P-MPJPE

Module	Time(s)	Params(M)	MPJPE↓	P-MPJPE↓
w/o CNN	121.87	9.64	40.98	29.44
w/o Spatail(1,2)	38.75	9.59	28.34	21.46
w/o Temporal(1,2)	74.77	1.3	28.84	21.96
w/o VE and TE	112.65	9.57	26.99	20.63

degree of improvement. However, after combining the spatiotemporal module, the performance reached its optimal level. Verified the rationality of our use of the spatiotemporal module in CTFM.

6) *Impact of Position Encoding*: Table VII shows the ablation experiment of position encoding required for the spatial module, temporal module, and viewpoint fusion. SE is the spatial position encoding, TE is the temporal position encoding, and VE is the viewpoint position encoding. These findings confirm the efficacy of our positional encoding strategy and further attest to the superior capabilities of the ECTFormer architecture.

7) *Impact of Number of Temporal Attention Layers*: Table VIII illustrates the impact of the number of layers in the temporal modeling strategy on overall performance. Using two conventional evaluation metrics, it demonstrates that the strategy achieves optimal results with a layer window of 4. We also provide the size of model parameters contributed by each layer to facilitate more precise comparisons.

8) *Impact of All Modules Throughout ECTFormer*: Table IX illustrates the impact of different modules on overall performance, employing both MPJPE and P-MPJPE metrics for evaluation, while also presenting the model parameters contributed by each module to the method. This experiment

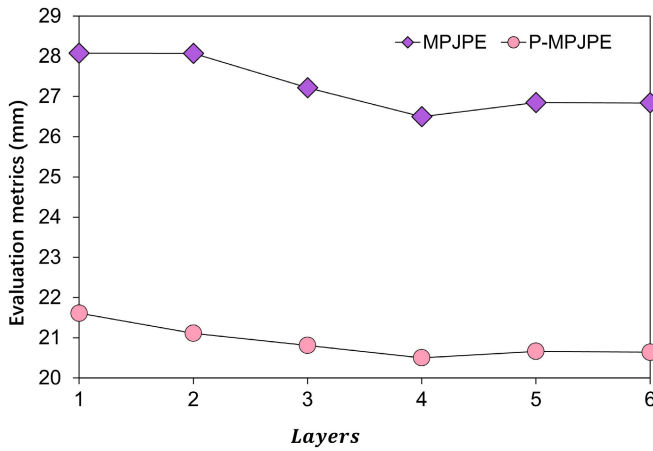


Fig. 7. Quantitative comparative experiments on the number of layers in spatiotemporal transformer models allowed us to visualize the impact of transformer depth on model accuracy, using MPJPE and PA-MPJPE as evaluation metrics. Results indicate that our method achieves optimal performance when employing a four-layer transformer configuration.



Fig. 8. Visualization of outdoor pose estimation results. To validate the method's performance, we selected an outdoor multiview 3-D human skiing pose dataset, Ski pose [49], which was not used during training, to demonstrate its superior capabilities.

validates the role of every module within our overall framework.

9) *Impact of Number of Transformer Block Layers in the Overall Method:* Fig. 7 shows the resulting plot for the quantity of block layers in the attention structure, which is divided into two metrics, MPJPE and P-MPJPE. The graph shows that the accuracy of our method is optimized when the number of attention layers used is 4. When the layer count surpasses four, the accuracy stabilizes and ceases to decline, confirming that this depth represents the optimal parameter configuration.

G. Real-World Scenario Validation

In Fig. 8, we selected an outdoor multiview 3-D human skiing pose dataset, Ski-Pose [49], not used in training, to demonstrate the superior performance of this method. In the figure, we chose four viewpoints and visualized the corresponding 2-D poses. Then, we visualized the predicted 3-D poses on the right.

H. Limitations

Despite ECTFormer's promising results, this study has several limitations that warrant acknowledgment. First, the model's performance can degrade under rapid human motion. The current hierarchical spatiotemporal module, while effective for capturing general cross-view correlations, may not fully resolve the severe motion blur and large displacement of fast-moving joints, leading to less accurate 3-D pose estimations in such scenarios. Second, the calibration-free design, though a significant advantage, inherently limits the framework's ability to leverage precise geometric constraints, which might be beneficial in controlled environments with available camera parameters. Finally, the generalizability of the model to unseen, highly occluded, or crowded real-world settings requires further validation.

Notwithstanding these limitations, we believe our work provides a solid foundation for future research. Subsequent efforts will focus on integrating more powerful motion priors or attention mechanisms to enhance temporal modeling for rapid movements and exploring self-supervised strategies to bridge the performance gap between calibration-free and calibration-based paradigms.

V. CONCLUSION

This article has introduced a novel calibration-free multiview 3-D HPE framework based on a hybrid CNN-Transformer architecture. The core of our approach lies in its hierarchical spatiotemporal feature extraction module, which effectively captures long-range cross-view correlations and geometric cues, and a computationally efficient fusion mechanism that optimally integrates multiview information. Extensive experimental validation confirms that the proposed method successfully mitigates the depth ambiguity inherent in monocular systems, achieves competitive performance without relying on precalibrated camera parameters, and maintains a compact model size, underscoring its practical efficiency.

The compelling advantages of our framework highlight its significant potential for application in complex, real-world scenarios. It is particularly suited for deployment in smart surveillance systems for multiperson tracking, athletic performance analysis in sports science, and interactive systems in entertainment and VR/AR, where the quick setup and invariance to camera calibration are critical.

Notwithstanding its strengths, we openly acknowledge the limitations of our work. Primarily, the model's performance can degrade when handling rapid and highly articulated human motions, as the current architecture may not fully resolve the severe ambiguities associated with fast-moving joints. Furthermore, the generalizability of our calibration-free approach to extremely crowded scenes or environments with heavy, persistent occlusions requires further in-depth investigation.

Future research will be directed toward overcoming these challenges. We plan to explore more powerful temporal modeling techniques, such as refined attention mechanisms, to enhance robustness to rapid motion. Concurrently, we will focus on developing advanced self-supervised learning

strategies to bridge the performance gap with calibration-based methods and to improve the framework's generalization in completely unconstrained real-world settings.

REFERENCES

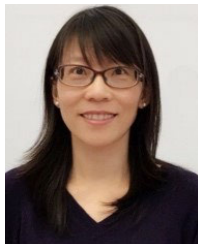
- [1] T.-H. Tsai and Y.-J. Luo, "Monocular-based 3-D human pose estimation with refinement block and special loss function," *IEEE Sensors J.*, vol. 25, no. 3, pp. 5679–5687, Feb. 2025, doi: [10.1109/JSEN.2024.3510728](https://doi.org/10.1109/JSEN.2024.3510728).
- [2] M. Zhang, Q. Gao, Y. Lai, J. Hu, X. Zhang, and Z. Ju, "HR-GCN: 2D–3D whole-body pose estimation with high-resolution graph convolutional network from a monocular camera," *IEEE Sensors J.*, early access, Apr. 10, 2025, doi: [10.1109/JSEN.2025.3557770](https://doi.org/10.1109/JSEN.2025.3557770).
- [3] H. He, G. Liu, X. Zhu, L. He, and G. Tian, "Interacting multiple model-based human pose estimation using a distributed 3D camera network," *IEEE Sensors J.*, vol. 19, no. 22, pp. 10584–10590, Nov. 2019, doi: [10.1109/JSEN.2019.2931603](https://doi.org/10.1109/JSEN.2019.2931603).
- [4] C. Zheng, S. Zhu, M. Mendieta, T. Yang, C. Chen, and Z. Ding, "3D human pose estimation with spatial and temporal transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 11656–11665.
- [5] J. Zhang, Z. Tu, J. Yang, Y. Chen, and J. Yuan, "MixSTE: Seq2seq mixed spatio-temporal encoder for 3D human pose estimation in video," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13232–13242.
- [6] S. Mehraban, V. Adeli, and B. Taati, "MotionAGFormer: Enhancing 3D human pose estimation with a transformer-GCNFormer network," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 6920–6930.
- [7] L. Chen and G. Wang, "CPFormer: End-to-end multi-person human pose estimation from raw radar cubes with transformers," *IEEE Sensors J.*, vol. 25, no. 7, pp. 12466–12478, Apr. 2025, doi: [10.1109/JSEN.2025.3542078](https://doi.org/10.1109/JSEN.2025.3542078).
- [8] L. Xu et al., "Group multi-view transformer for 3D shape analysis with spatial encoding," *IEEE Trans. Multimedia*, vol. 26, pp. 9450–9463, 2024.
- [9] G. Hua, H. Liu, W. Li, Q. Zhang, R. Ding, and X. Xu, "Weakly-supervised 3D human pose estimation with cross-view U-shaped graph convolutional network," *IEEE Trans. Multimedia*, vol. 25, pp. 1832–1843, 2023.
- [10] Y. Gao and S. Wang, "Multi-view human pose estimation via temporal fusion with attention adaptive fusion weights," *IEEE Sensors J.*, early access, May 21, 2025, doi: [10.1109/JSEN.2025.3570373](https://doi.org/10.1109/JSEN.2025.3570373).
- [11] K. Isakov, E. Burkov, V. Lempitsky, and Y. Malkov, "Learnable triangulation of human pose," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Jan. 2019, pp. 7718–7727.
- [12] J. Lin and G. H. Lee, "Multi-view multi-person 3D pose estimation with plane sweep stereo," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Aug. 2021, pp. 11886–11895.
- [13] B. Yang, H. Wang, L. Hu, H. Zhu, C.-T. Lam, and K. Fang, "Few-shot cross-domain-based WiFi sensing system for online learning in IoT," *IEEE Sensors J.*, vol. 23, no. 23, pp. 29892–29905, Dec. 2023.
- [14] J. Song, C. Pun, H. Li, R. Lan, J.-C. Xie, and H. Gao, "Local optimization networks for multi-view multi-person human posture estimation," in *Proc. IEEE Int. Conf. Acoust., Aug. 2024*, pp. 3995–3999.
- [15] B. Gordon, S. Raab, G. Azov, R. Giryas, and D. Cohen-Or, "FLEX: Extrinsic parameters-free multi-view 3D human motion reconstruction," in *Proc. 17th Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2022, pp. 176–196.
- [16] L. Zhang et al., "Progressive multi-view fusion for 3D human pose estimation," in *Proc. IEEE Int. Conf. Image Process.*, Apr. 2023, pp. 1600–1604.
- [17] J. Guo et al., "EquArchNets: Equivariant neural networks-based approach for foot arch classification toward variations in foot orientation," *IEEE Sensors J.*, vol. 25, no. 8, pp. 14328–14341, Apr. 2025.
- [18] L. Zhang, K. Zhou, F. Lu, X. Zhou, and Y. Shi, "Deep semantic graph transformer for multi-view 3D human pose estimation," in *Proc. AAAI Conf. Artif. Intell.*, vol. 38, 2024, pp. 7205–7214.
- [19] L. Zhao, X. Peng, Y. Tian, M. Kapadia, and D. N. Metaxas, "Semantic graph convolutional networks for 3D human pose regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3425–3435.
- [20] S. Yan, Y. Xiong, and D. Lin, "Spatial temporal graph convolutional networks for skeleton-based action recognition," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, 2018, pp. 7444–7452.
- [21] B. X. B. Yu, Z. Zhang, Y. Liu, S.-H. Zhong, Y. Liu, and C. W. Chen, "GLA-GCN: Global-local adaptive graph convolutional network for 3D human pose estimation from monocular video," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 8784–8795.
- [22] W. Li et al., "GraphMLP: A graph MLP-like architecture for 3D human pose estimation," *Pattern Recognit.*, vol. 158, Feb. 2025, Art. no. 110925.
- [23] K. Gong, J. Zhang, and J. Feng, "PoseAug: A differentiable pose augmentation framework for 3D human pose estimation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 8575–8584.
- [24] W. Shan, Z. Liu, X. Zhang, S. Wang, S. Ma, and W. Gao, "P-STMO: Pre-trained spatial temporal many-to-one model for 3D human pose estimation," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 461–478.
- [25] Z. Tang, Z. Qiu, Y. Hao, R. Hong, and T. Yao, "3D human pose estimation with spatio-temporal criss-cross attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 4790–4799.
- [26] W. Li, H. Liu, H. Tang, P. Wang, and L. Van Gool, "MHFormer: Multi-hypothesis transformer for 3D human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13147–13156.
- [27] W. Zhu, X. Ma, Z. Liu, L. Liu, W. Wu, and Y. Wang, "MotionBERT: A unified perspective on learning human motion representations," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 15039–15053.
- [28] Z. Sun et al., "Repose: 3D human pose estimation via spatio-temporal depth relational consistency," in *Proc. Eur. Conf. Comput. Vis.*, May 2024, pp. 309–325.
- [29] Y. Huang, J. Liu, K. Xian, and R. C. Qiu, "PoseMamba: Monocular 3D human pose estimation with bidirectional global-local spatio-temporal state space model," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, 2025, pp. 3842–3850.
- [30] X. Zhang, Q. Bao, Q. Cui, W. Yang, and Q. Liao, "Pose magic: Efficient and temporally consistent human pose estimation with a hybrid mamba-GCN network," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, 2025, pp. 10248–10256.
- [31] E. Remelli, S. Han, S. Honari, P. Fua, and R. Wang, "Lightweight multi-view 3D pose estimation through camera-disentangled representation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6039–6048.
- [32] Z. Liao, J. Zhu, C. Wang, H. Hu, and S. L. Waslander, "Multiple view geometry transformers for 3D human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 708–717.
- [33] T. Wang, J. Zhang, Y. Cai, S. Yan, and J. Feng, "Direct multi-view multi-person 3D pose estimation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 13153–13164.
- [34] H. Tu, C. Wang, and W. Zeng, "VoxelPose: Towards multi-camera 3D human pose estimation in wild environment," in *Proc. Eur. Conf. Comput. Vis.*, Aug. 2020, pp. 197–212.
- [35] H. Ye, W. Zhu, C. Wang, R. Wu, and Y. Wang, "Faster VoxelPose: Real-time 3D human pose estimation by orthographic projection," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 142–159.
- [36] B. Jiang, L. Hu, and S. Xia, "Probabilistic triangulation for uncalibrated multi-view 3D human pose estimation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 14804–14814.
- [37] L. Zhang et al., "ESMformer: Error-aware self-supervised transformer for multi-view 3D human pose estimation," *Pattern Recognit.*, vol. 158, Feb. 2025, Art. no. 110955.
- [38] Y. Chen, Z. Wang, Y. Peng, Z. Zhang, G. Yu, and J. Sun, "Cascaded pyramid network for multi-person pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7103–7112.
- [39] W. Shan et al., "Diffusion-based 3D human pose estimation with multi-hypothesis aggregation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 14715–14725.
- [40] J. Peng, Y. Zhou, and P. Y. Mok, "KTPFormer: Kinematics and trajectory prior knowledge-enhanced transformer for 3D human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 1123–1132.
- [41] D. C. Luvizon, D. Picard, and H. Tabia, "Consensus-based optimization for 3D human pose estimation in camera coordinates," *Int. J. Comput. Vis.*, vol. 130, no. 3, pp. 869–882, Mar. 2022.
- [42] S. Bultmann and S. Behnke, "Real-time multi-view 3D human pose estimation using semantic feedback to smart edge sensors," 2021, *arXiv:2106.14729*.

- [43] H. Qiu, C. Wang, J. Wang, N. Wang, and W. Zeng, "Cross view fusion for 3D human pose estimation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jul. 2019, pp. 4342–4351.
- [44] C. Ionescu, D. Papava, V. Olaru, and C. Sminchisescu, "Human3.6M: Large scale datasets and predictive methods for 3D human sensing in natural environments," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 7, pp. 1325–1339, Jul. 2014.
- [45] D. Mehta et al., "Monocular 3D human pose estimation in the wild using improved CNN supervision," in *Proc. Int. Conf. 3D Vis. (3DV)*, Oct. 2017, pp. 506–516.
- [46] M. Gholami, A. Rezaei, H. Rhodin, R. Ward, and Z. J. Wang, "Self-supervised 3D human pose estimation from video," *Neurocomputing*, vol. 488, pp. 97–106, Jun. 2022.
- [47] B. Wandt, M. Rudolph, P. Zell, H. Rhodin, and B. Rosenhahn, "Canon-Pose: Self-supervised monocular 3D human pose estimation in the wild," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13289–13299.
- [48] K. Zhou, L. Zhang, F. Lu, X.-D. Zhou, and Y. Shi, "Efficient hierarchical multi-view fusion transformer for 3D human pose estimation," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 7512–7520.
- [49] B. Fasel et al., "Three-dimensional body and centre of mass kinematics in Alpine ski racing using differential GNSS and inertial sensors," *Remote Sens.*, vol. 8, no. 8, p. 671, Aug. 2016.



Jucheng Song received the M.Sc. degree in electronic information from Nanjing University of Posts and Telecommunications, Nanjing, China, in 2024. He is currently pursuing the Ph.D. degree with the School of Computer Applications, Macao Polytechnic University (MPU), Macau, China.

His current research interests include computer vision, human motion analysis, and human–computer interaction.



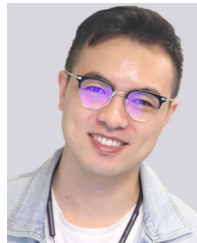
Xu Yang (Member, IEEE) received the B.Sc. degree from the University of Electronic Science and Technology of China, Chengdu, China, in 1997, and the M.Sc. and Ph.D. degrees from the Queen Mary University of London, London, U.K., in 2003 and 2009, respectively.

She joined the Faculty of Applied Sciences, Macao Polytechnic University (MPU), Macau, China, in 2013 as a Lecturer. Her current research interests include wireless communications, computer vision, medical image analyses, and machine learning.



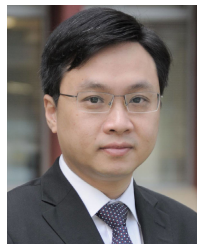
Yapeng Wang (Member, IEEE) received the B.Eng. degree in telecommunication engineering and the B.Sc. degree in computer and its applications from North China Electric Power University, Beijing, China, in 1998, and the M.Sc. degree in internet computing and the Ph.D. degree in electronic engineering from the Queen Mary University of London, London, U.K., in 2002 and 2007, respectively.

He joined the Faculty of Applied Sciences, Macao Polytechnic University (MPU), Macau, China, in 2021, as an Associate Professor. His current research interests include applied artificial intelligence, wireless communications, automatic speech recognition, nature language processing, medical image analysis, and machine learning.



Jie Zhang (Member, IEEE) received the Ph.D. degree from The Hong Kong Polytechnic University, Kowloon, Hong Kong, in 2023.

He is currently a Lecturer with the Faculty of Applied Sciences, Macao Polytechnic University (MPU), Macau, SAR, China. He has published over 40 peer-reviewed journal articles (including *IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS (TVCG)*, *IEEE TRANSACTIONS ON MULTIMEDIA (TMM)*, *IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT (TIM)*, *CAD*, *IJHCS*, *IJHCI*), three textbooks, patents and international conference papers. His current research interests include AI for design and graphics, 3-D generative models, color science, data visualization, ergonomic design, human–computer interaction, and 3-D Anatomical human modeling/editing.



Sio-Kei Im (Senior Member, IEEE) received the B.Sc. degree in computer science and the M.Sc. degree in enterprise information system from the King's College, University of London, London, U.K., in 1998 and 1999, respectively, and the Ph.D. degree in electronic engineering from the Queen Mary University of London (QMUL), London, in 2007.

He gained the position of a Lecturer within the Computing Programme, Macao Polytechnic University (MPU), Macau, China, in 2001. In 2005, he became the Operations Manager of the MPU-QMUL information systems research center jointly operated by MPU and QMUL, where he carried out signal processing work. He was promoted to a Professor with Macao Polytechnic University in 2015. He was a Visiting Scholar with the School of Engineering, University of California at Los Angeles (UCLA), Los Angeles, CA, USA; and an Honorary Professor at The Open University of Hong Kong, Hong Kong.